Deep Ensembles for Predictive Uncertainty

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1 Introduction

Neural network classifier performance is typically evaluated by its accuracy, however in many scenarios the probabilities a network assigns to its predictions matters as much as the predictions and raw accuracy. It is frequently observed that a model's assigned probability is a poor descriptor to a model's true confidence of a given prediction (i.e. when a model is poorly calibrated). A simple approach to provide insight into model uncertainty is through ensembling, by averaging the predictions of several models, the ensemble becomes less confident when the members disagree. In this report we compare two different methods of ensembling. A standard deep ensemble which consists of several networks trained separately from different random initializations, and MC dropout where a single network with dropout is trained and each stochastic forward pass at test time represents a single member of the ensemble. We first evaluate both models on a simple two-dimensional problem where predictive uncertainty can be visualized directly and then evaluate them on Fashion-MNIST. For each method we examine the full ensemble and its individual members with respect to accuracy, predictive uncertainty and calibration, as well as the diversity of their predictions.

2 Setup

2.1 Data

[Two moons: 500 train / 1,000 test points, noise 0.12; 2-D, so predictions can be plotted over the whole input plane. Fashion-MNIST: 60,000 train / 10,000 test 28×28 grayscale images, 10 classes; several classes look alike (shirt, T-shirt, pullover, coat), which gives real aleatoric uncertainty.]

2.2 Model and training

For both of the following methods (standard deep ensembles and MC dropout), we use the same MLP architecture, which consists of a flattening layer, two hidden ReLU layers of width 64 for two moons and 256 for Fashion-MNIST, and a linear output layer. We use Adam's optimizer with a learning rate of 10^{-3}, and train for 200 epochs on two moons and 20 epochs on Fashion-MNIST with a batch size of 128 and 256, respectively. The training loss is the cross-entropy on the output logits. Our L2 regularization term is set to a weight decay of 10^{-4}, which corresponds to a zero-mean isotropic Gaussian prior on the weights. The only differences between the two methods are the use of dropout and how the members of the ensemble are produced.

2.3 Standard deep ensembles

The ensemble prediction averages the members' predicted probabilities,

$$\bar{p}(y \mid x) = \frac{1}{M} \sum_{m=1}^{M} p_m(y \mid x), \tag{1}$$

an empirical approximation of the predictive distribution $p(y \mid x, \mathcal{D})$.

For our implementation, we use $M = 10$ networks, where each member m is seeded with `seed`$+m$ to ensure different initializations. No dropout is used in this method, because that is the comparison method we are evaluating against. Each member is trained on the same full training set, and the final ensemble prediction is the average of the 10 members' predictions, see Eq. (1).

2.4 MC dropout

[One network with dropout $p = 0.1$ after each hidden layer, trained with the same objective (approximate variational inference, Gal & Ghahramani). Dropout stays on at test time; $T = 10$ stochastic passes are each treated as a member and averaged as in Eq. (1). Masks use a fixed seed, so results are reproducible.]

2.5 Evaluation metrics

All metrics except the entropy decomposition and disagreement are computed on $\bar{p}$. We use the following metrics to evaluate the performance of the models:

- **Error rate**: fraction of inputs where $\arg\max_c \bar{p}(c \mid x) \neq y$.
- **Negative log-likelihood (NLL)** (nats): $\mathrm{NLL} = -\frac{1}{N}\sum_n \ln \bar{p}(y_n \mid x_n)$.
- **Expected calibration error (ECE)**: with confidence $\max_c \bar{p}(c \mid x)$ split into $B = 15$ equal-width bins, $\mathrm{ECE} = \sum_b \frac{n_b}{N} |\mathrm{acc}(b) - \mathrm{conf}(b)|$, shown in full by reliability diagrams.
- **Entropy decomposition**: predictive entropy (total) splits into expected entropy (aleatoric) and mutual information (epistemic),

$$\underbrace{H[\bar{p}]}_{\text{predictive}} = \underbrace{\frac{1}{M}\sum_m H[p_m]}_{\text{expected}} + \underbrace{\mathrm{MI}}_{\text{epistemic}}, \qquad H[p] = -\sum_c p_c \ln p_c. \tag{2}$$

- **Pairwise disagreement**: fraction of test inputs on which two members predict a different class; a PCA of each member's test predictions (*function space*) shows the same diversity visually.

2.6 Ensemble-size protocol

[Train $M = 10$ once; evaluate subsets of size $k \in \{1, 2, 5, 10\}$ (every subset when there are at most 20, else 20 random ones). Error bars are ± 1 sd over subsets; they shrink at large k partly because subsets overlap, and $k = 10$ is a single subset. The test set is used for evaluation only, never for model selection.]

3 Evaluation

Figures show the deep ensemble (a) and MC dropout (b) side by side; each panel has its own y-axis range, so compare values rather than curve heights.

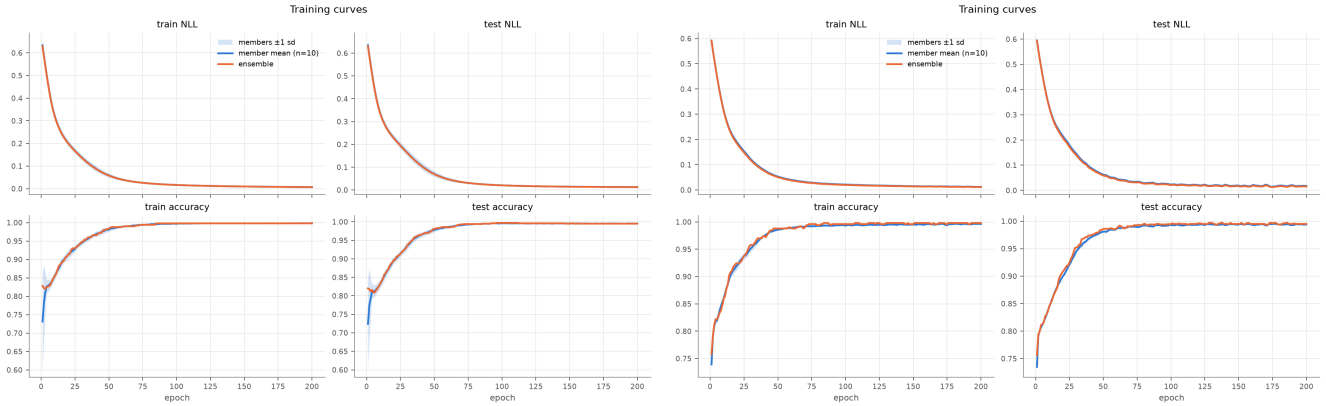
3.1 Two moons

3.1.1 Training curves

Both the Deep Ensemble and MC Dropout methods are trained on the two moons dataset, and their training and testing curves (including NLL and accuracy) are shown in Figure 1.

We observe that they both converge by epoch 75, achieving a test accuracy of 99.3%. Their test NLL continues to decrease through training, reaching about 0.013 (DE) and 0.015 (MC Dropout) at epoch 200. We note that the training and testing accuracies both remain close, without any meaningful loss in test accuracy later in training. A similar observation is true for the NLL, which continues to decrease on the test set without any sustained increase. Thus, we conclude that there is no evidence of overfitting in either method. The final gain from ensembling 10 members is negligible; the Deep Ensemble method has a final gain of -0.04% percentage points, and the MC Dropout method has a final gain of $+0.07\%$ percentage points.

The ensemble and member-mean curves become very similar after the early training phase (about epoch 10), which is consistent with the negligible final gain from averaging on this simple dataset. Member diversity is examined more directly using mutual information and pairwise disagreement in Section 3.1.2 and Section 3.1.4, respectively.



(a) Deep ensemble (b) MC dropout

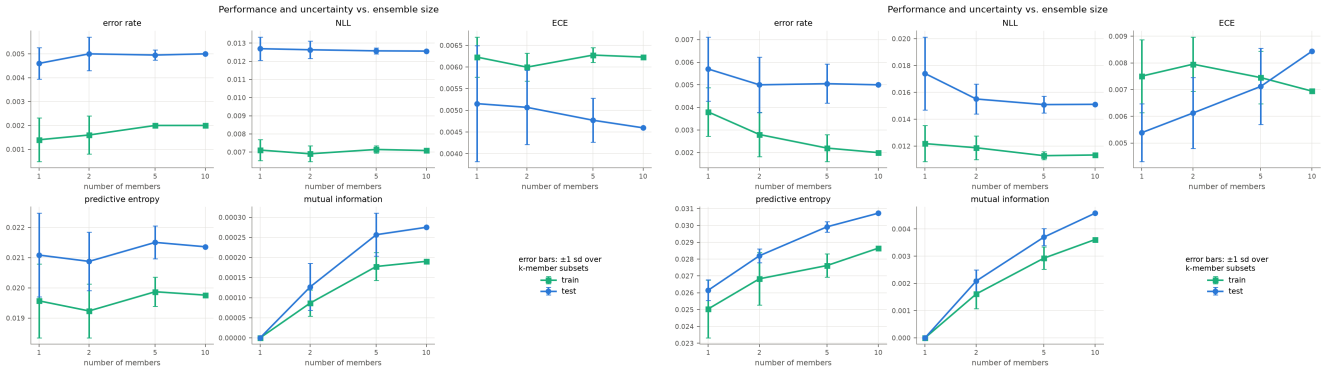
Figure 1: Two moons training curves: member mean ±1 sd and the averaged prediction, on train and test.

3.1.2 Ensemble size

We first observe that the Deep Ensemble method does not meaningfully benefit from increasing the number of members, unlike the MC dropout method, which shows some improvement. We can see this in Figure 2, where the test error, NLL, ECE, and MI are plotted against the number of members k, where $k \in \{1, 2, 5, 10\}$. The Deep Ensemble method has close to flat curves for most metrics, with some small fluctuations (Figure 2a). This means that the Deep Ensemble method does not benefit from increasing the number of members, and that the members are nearly identical on the test points. The mutual information curve does increase, but only by 0.0003 nats, so it is negligible. Predictive entropy is unchanged for the Deep Ensemble method, however it does increase a little for the MC dropout method.

MC dropout also shows some improvement in training error and NLL as k increases, while its test error remains approximately constant at 0.5%. Test NLL decreases slightly from 0.0174 for one member to 0.0151 for ten members. In contrast, test ECE increases a little, indicating slightly worse calibration after averaging in this setting; this small change is not enough to be considered evidence of overfitting. Mutual information increases to 0.0046 nats at $k = 10$, substantially above the deep ensemble's 0.0003 nats, indicating greater diversity among MC-dropout predictions.

We note that all aforementioned effects are very small because the one member predictions are already capable of solving this simple task.



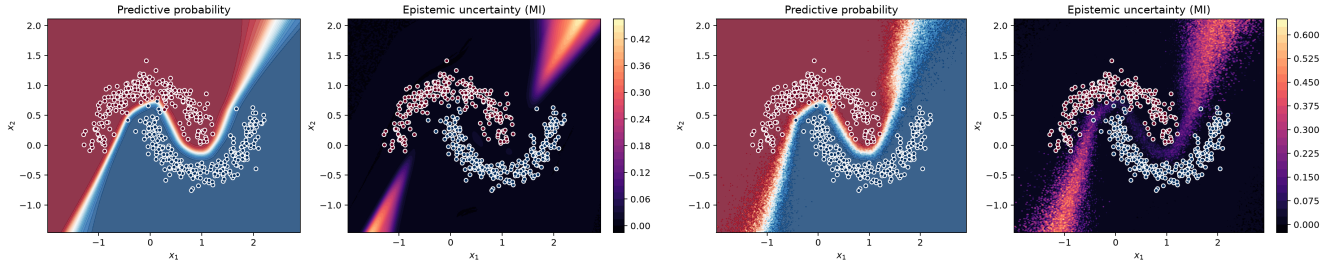
(a) Deep ensemble (b) MC dropout

Figure 2: Two moons metrics vs. number of members; error bars are ±1 sd over k-member subsets. Note the very small y-axis ranges.

3.1.3 Uncertainty surfaces

We observe that both methods draw a clear and correct decision boundary between the two moons; see in Figure 3. They also show low epistemic uncertainty (mutual information) near the data (just above 0.003 for DE and 0.022

for MC Dropout), and higher uncertainty away from the data by a factor of 12 for Deep Ensemble and 2.4 for MC Dropout. The MC Dropout method also has a higher maximum mutual information (0.053) than the Deep Ensemble method (0.038), and more uncertainty spread between the two moons, including on the data itself.



(a) Deep ensemble (b) MC dropout

Figure 3: Two moons: predictive probability (left of each pair) and mutual information (right) over the input plane.

3.1.4 Function space

With respect to the subfigures in Figure 4, we first notice that the PCA of test predictions (the full probability vectors) look quite different between the two models. The deep ensemble members are very tightly clustered, while the MC dropout members are more spread out. This is likely due to the random dropout mask, which masks hidden-unit activations, so this evaluates different subnetworks each time. You can also see clear separation between the two methods themselves, which means that the methods make different predictions overall.

We also note that MC dropout has a higher mean pairwise disagreement (0.5%) than the deep ensemble (0.1%), which is consistent with the PCA plot and our expectations of both models. There are only 4 test points that change class under any deep ensemble member, while there are about 14 boundary points that change class for at least one MC dropout model. Though there is a larger spread in the MC dropout probability space, both methods agree on the majority of the test points. This aligns with our earlier conclusion that neither method benefits very much from increasing the number of members.

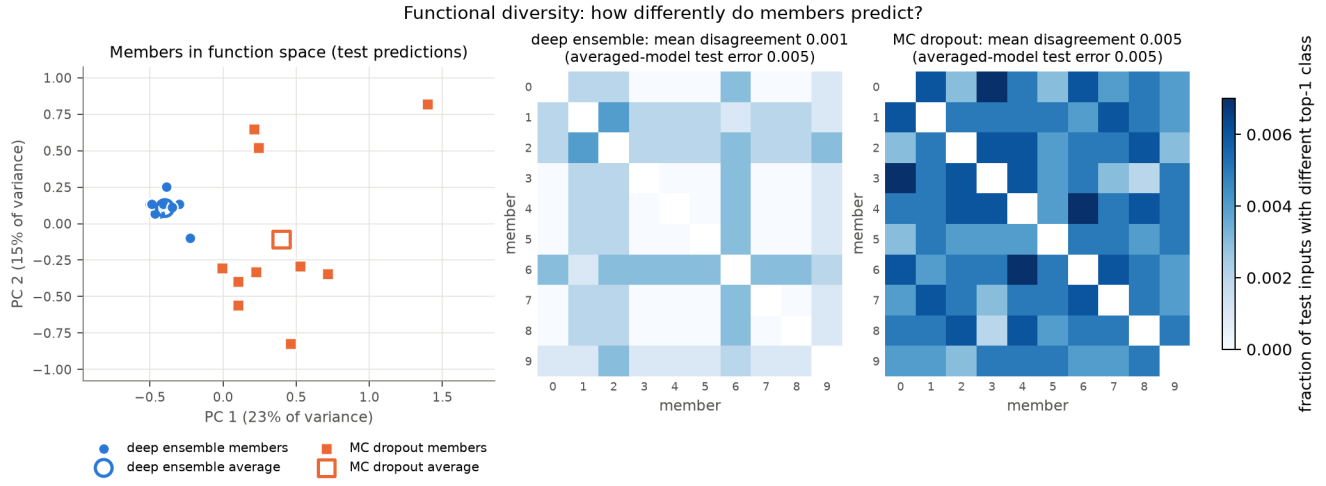
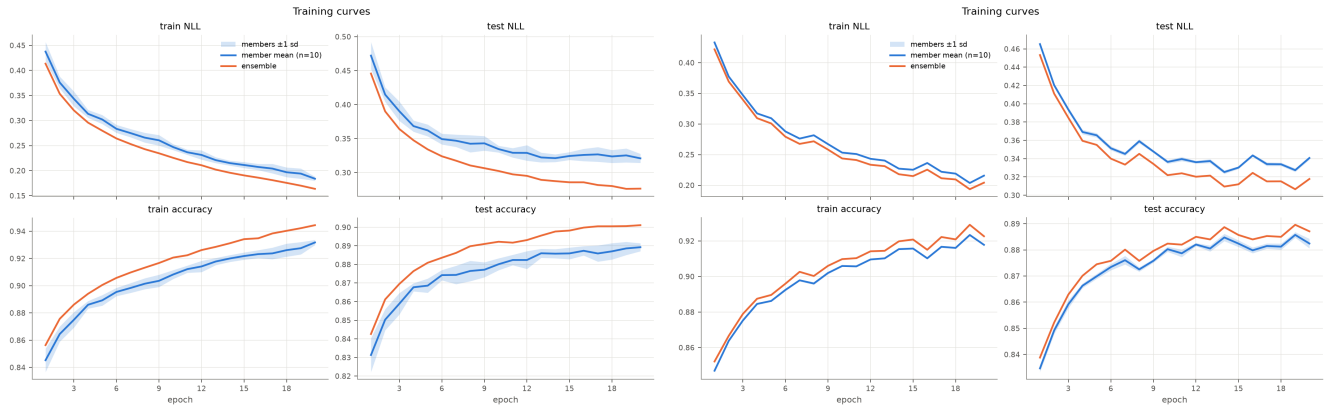


Figure 4: Two moons function space: members of both methods in one PCA of their test predictions (left) and pairwise top-1 disagreement (right).

3.2 Fashion-MNIST

3.2.1 Training curves

[DE 90.1% vs. MC 88.7% test accuracy; test NLL 0.276 vs. 0.318. Gain over the member mean: DE up to +1.46% (final +1.18%), MC +0.47%. Member spread is larger for the DE in NLL (sd 0.0067 vs. 0.0018) but similar in accuracy; the bands mean different things (10 networks vs. 10 passes of one). Train–test accuracy gap: DE 4.3 points, MC 3.6 points (dropout regularizes).]

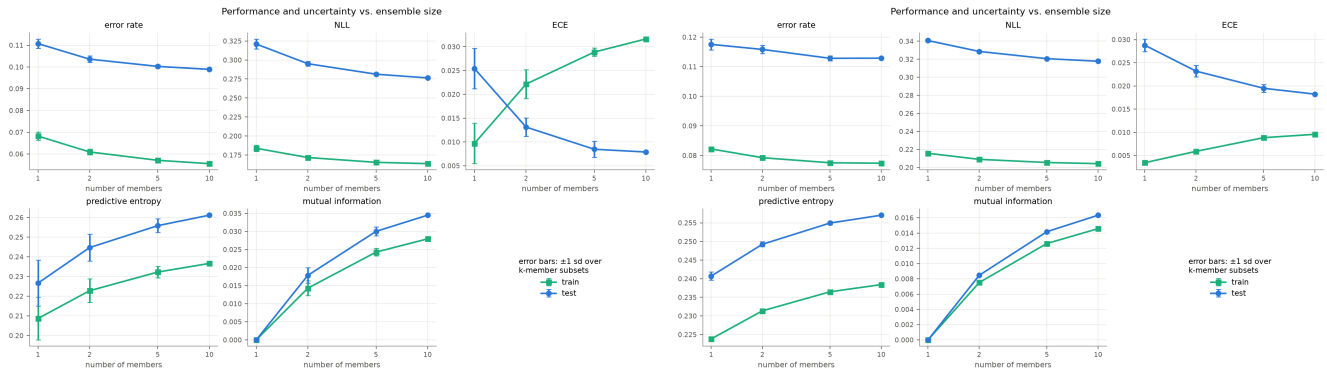


(a) Deep ensemble (b) MC dropout

Figure 5: Fashion-MNIST training curves: member mean ±1 sd and the averaged prediction, on train and test.

3.2.2 Ensemble size

[Every metric improves with k, more for the DE (error 11.1%→9.9% vs. 11.8%→11.3%; ECE 0.025→0.008 vs. 0.029→0.018; MI 0.035 vs. 0.016 at $k = 10$); most of the gain comes by $k = 5$. Context: uniform-guess NLL and maximum entropy are $\ln 10 = 2.30$; MI is about 13% of predictive entropy for the DE. Train ECE rises with k while test ECE falls: averaging lowers confidence, which fixes overconfident test predictions but makes already slightly underconfident train predictions more so.]



(a) Deep ensemble (b) MC dropout

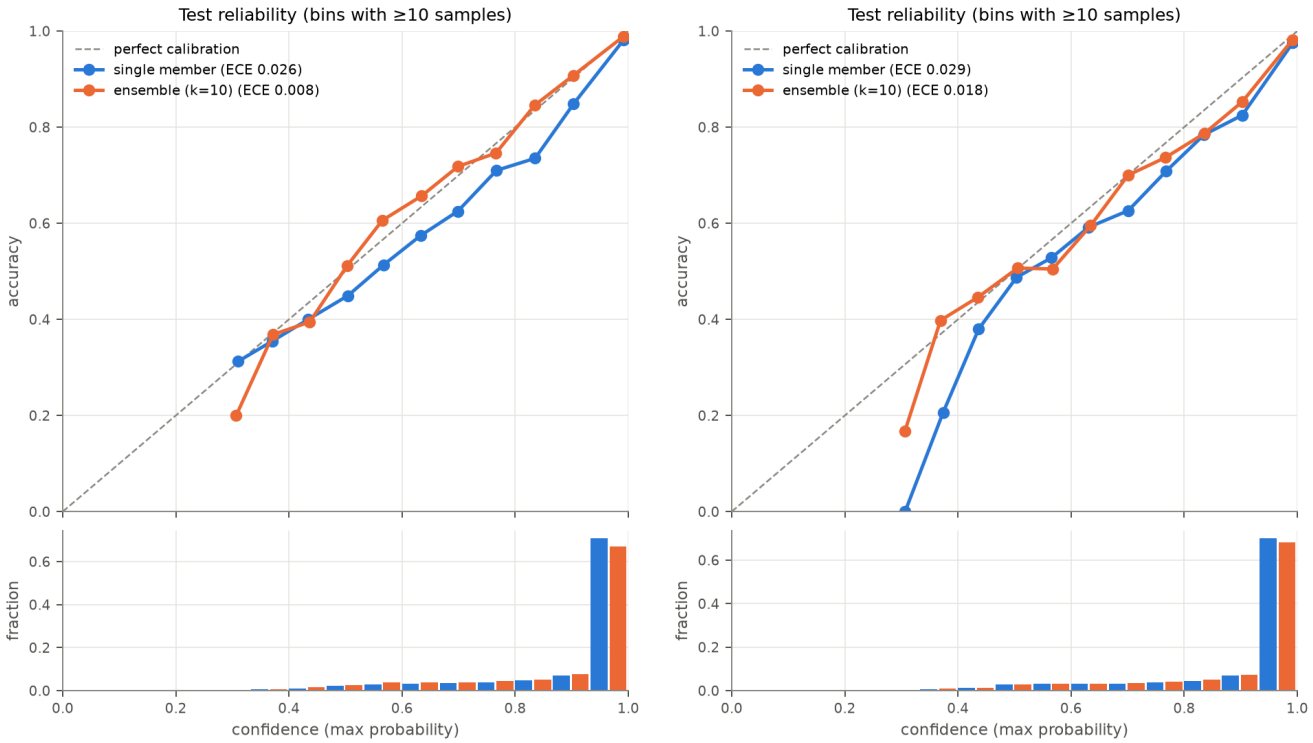
Figure 6: Fashion-MNIST metrics vs. number of members; error bars are ±1 sd over k-member subsets.

3.2.3 Reliability

[Single members of both methods are overconfident (mean confidence − accuracy: DE +2.6, MC +2.9 points). The DE ensemble ends almost exactly calibrated (−0.06 points, ECE −69%); MC dropout stays overconfident (+1.7 points, ECE −38%). Averaging lowers confidence most where members disagree, which is where errors concentrate.]

3.2.4 Function space

[DE members spread widely; the 10 MC passes almost coincide. Mean disagreement DE 7.1% vs. MC 4.4%, matching the MI gap. PC1 mostly separates the methods (they trained different networks); diversity is the spread within each group. Contrast with two moons (Fig. 4), where the order is reversed.]



(a) Deep ensemble

(b) MC dropout

Figure 7: Fashion-MNIST test reliability: single member vs. full ensemble. Points below the diagonal are overconfident; the histogram shows each bin's share of inputs.

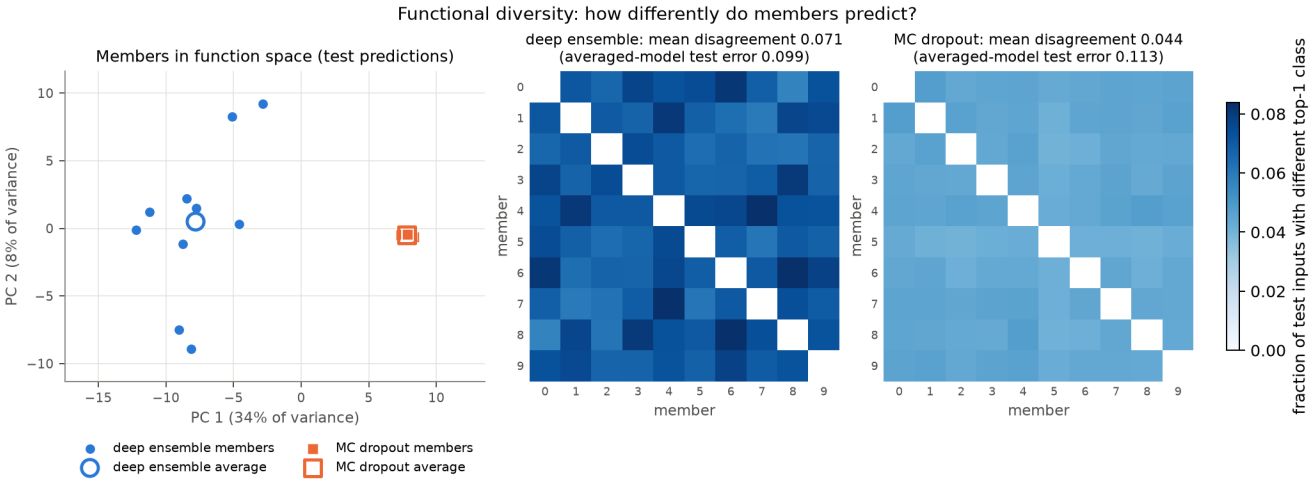


Figure 8: Fashion-MNIST function space: members of both methods in one PCA of their test predictions (left) and pairwise top-1 disagreement (right).

4 Discussion

[Where diversity comes from: random initialization sends DE members to different solutions on both datasets; what matters is whether the test inputs force them to agree. On two moons (2-D, test points on the training data) every solution draws the same boundary; on Fashion-MNIST (784-D, ambiguous images) they differ. MC dropout samples around one solution on both datasets (Fort et al., 2019).]

[The same averaging fixes calibration on test and worsens it on train.]

[Cost: the DE trains 10 networks; MC dropout trains one and pays only at test time.]

[Limitations: one training seed per configuration; dropout rate 0.1 not tuned; MLPs only; no out-of-distribution or dataset-shift evaluation.]

5 Conclusion

[Both methods improve on a single network, and the improvement grows with k. The DE improves more on accuracy, NLL and especially calibration, because its members differ more ($2\times$ MI, $1.6\times$ disagreement). In-distribution diversity depends on whether the data leaves room for it; on two moons it does not.]